

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I M. Arshini (192424322) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Arshini

90

150

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Artificial Intelligence.

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Descriptive test

Uniform Cost Search (UCS):-

- * Explore the path with the lowest cumulative cost first
- * finds the optimal (least-cost) path to the exist
- * Suitable when different paths have different movements costs

Advantages:-

- * Complete (finds a solution if its exist)
- * Optimal for positive path costs.

Disadvantages:-

- * High memory usage because it stores many nodes
- * can be slow for large mazes.

Working:-

- * Start from initial position
- * Expand the node with the minimum path cost
- * continue until the exist is reached.
- * Always finds the optimal path when path costs are positive.

Complexity:-

* Time complexity $O(b^{1 + \lceil \frac{C^*}{\epsilon} \rceil})$

* Space complexity: $O(b^{1 + \lceil \frac{C^*}{\epsilon} \rceil})$

where

b = branching factor

C^* = optimal solution cost

ϵ = minimum step cost.

a) Iterative Deepening Search (IDS):-
IDS repeatedly performs depth-first search with increasing depth limits until the goal is found.

Working:-

* Search depth = 0

* If goal not found, increases depth to 1

* Continue increasing depth until the exist is reached.

Advantages:-

* Uses very little memory

* Complete algorithm

* Finds the shallowest solution.

Disadvantages:-

- * Repeats nodes expansion
- * Does not guarantee the least-cost path.

Complexity:-

- * Time :: $O(b^d)$
- * Space :: $O(bd)$

where:-

b = branching factor

d = depth of solution.

3) Intelligent Agent types

* Simple Reflex Agent:-

- * Makes decisions only from current state
- * cannot remember previous locations
- * not suitable for maze solving.

Model-based Agent:-

- * Maintains an internal map of explored paths
- * Avoids revisiting the same locations.

Goal Based Agents:

- * searches for the exit using search algorithms
- * plans actions to reach the goal

Utility Based Agent:

- * Chooses the safest or shortest path based on utility.

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a) A smart traffic management system uses a Artificial Intelligence to control traffic signals based on real-time traffic conditions. The system collect data from sensors and cameras and adjusts signal timings to reduce congestion and improve the road safety.

1) simple reflex agent:

Working :-

* If the number of vehicle on one road increases, the signal immediately turns green.

* If pedestrians press the crossing button the pedestrian signal becomes active

* If there are no vehicles, the signal changes according to default timing.

Advantages:-

- * very fast response
- * easy to design and implement
- * low computational cost.

Disadvantages:-

- * cannot remember previous traffic conditions
- * cannot predict future traffic
- * cannot handle complex situations like accidents or roadblocks.

2) Model-based reflex Agent:-

It stores previous traffic information and combines it with current sensor inputs to make better decisions

Working:-

- * stores previous traffic density
- * maintains signal history
- * Detects repeated congestion
- * Predicts nearby road condition.

Advantages:-

- * Handles partially observable
- * makes more accurate decisions.

- * prevents unnecessary signal switching
- * improves traffic coordination.

Disadvantages:-

- * Requires more memory
- * more complex than simple reflex agent
- * higher computational cost

2) Goal-Based Agent :-

- * Reduce traffic congestion
- * Reduce waiting time
- * Ensure pedestrian safety
- * Allow emergency vehicles to pass quickly.
- * minimizes the fuel consumption
- * Improves traffic flow.

Working:-

The system evaluates multiple signal timings options and chooses the one that best achieves the desired objective.

Advantages:-

Intelligent planning

Better decision - making

Flexible and adaptable

Suitable for large smart cities

Disadvantages

* Requires more computation

* Decision - making takes more time

4) Utility - Based agent:-

evaluates every possible actions using a utility function and selects the actions that provides the highest overall benefit.

The utility function considers factors such as:

* vehicle waiting time

* Traffic density

* fuel consumption

* Emergency vehicle priority

* Pedestrian waiting time

* weather condition.

working:-

Instead of selecting the first possible action & solution, the system compares different signal timings and chooses the one that gives maximum overall efficiency.

Advantages:-

- * produces optimized decisions
- * Balances multiple objectives
- * minimizes congestion
- * Reduces pollution and fuel consumption.

Disadvantages

- * produces optimized decisions
- * minimizes congestion.

2) Learning Agent:-

Learning element, critic, problem generator

working:-

Peak traffic hours

Rainy season traffic

Accident-prone areas.

Advantages

- * Learn continuously
- * Improves accuracy over time.

Disadvantages

- * Requires large datasets
- * Training takes time
- * more expensive to develop.